

VoIP / Dialpad Troubleshooting Guide

Call quality and telephony issue resolution

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Call Quality Fundamentals

Voice needs three things: bandwidth (~100 Kbps per call, trivial), low latency (<150ms one-way), and low jitter/loss (<30ms jitter, <1% loss). Nearly all VoIP problems are network problems wearing a costume. Get evidence first: the platform's per-call quality analytics show which leg (user network, ISP, carrier) degraded.

Symptom Mapping

Symptom	Usual Cause	Fix
Choppy/robotic audio	Packet loss or jitter on the user's network	Wired > WiFi for heavy phone users; QoS/DSCP prioritization on firewall; check for saturation (backups running mid-day?)
One-way audio	NAT/firewall issue: SIP ALG mangling packets	DISABLE SIP ALG on the router/firewall (fixes a huge share of VoIP tickets); verify required ports/consistent NAT
Calls drop at exact intervals (e.g., 15 min)	Session timer / UDP timeout on firewall	Increase UDP session timeout; enable keep-alives
Echo	Analog leg or speakerphone acoustics	Headset test isolates it; replace analog adapters; reduce speaker volume
Registration failures (softphone offline)	Local internet, VPN interference, or expired sign-in	Re-auth the app; split-tunnel VoIP around VPN; verify no double-NAT
Poor quality only on WiFi calling	Roaming, co-channel interference, power-save on the NIC	See WiFi section of BVT-RUN-058; disable NIC power saving for heavy users

Dialpad-Specific Checks

- Status page first for platform-wide reports before deep local diagnosis.
- Per-user: correct audio device selected in app settings (docks/headsets hijack defaults); app version current; try web app to isolate desktop-app issues.
- Admin console: user license/role, call routing rules and business hours (the 'missed calls' ticket is usually routing config, not outage), voicemail and IVR assignments.
- Number porting issues: verify port order status and LOA accuracy; inbound failures post-port are usually carrier-side routing — open a ticket with trace examples (timestamp, caller, callee).

Site Readiness Standard (New Deployments)

- QoS policy marking voice traffic end-to-end on managed networks.

- SIP ALG disabled at the edge; consistent NAT; documented in BVT-OPS-026.
- Wired drops for reception/heavy call roles; tested headsets from the approved list.
- Pre-deployment network assessment (jitter/loss test to the provider) recorded for baseline.